

Generalizability of Sentiment-Based Stock Market Prediction

A replication study testing whether a classical logistic-regression GLM can predict daily US market direction from news sentiment

Can today's news headlines predict whether the stock market goes up tomorrow?

This question sits at the intersection of natural-language processing and financial markets. If public news sentiment predicts daily market direction, it challenges the idea that markets process information efficiently and opens the door to systematic trading strategies.

A classical logistic regression beat FinBERT and GPT-4 on Nigerian financial news

Shobayo et al. (2024) classified daily moves of the Nigerian NGX All-Share Index from news headlines by benchmarking FinBERT and GPT-4 against a simple **TF-IDF + logistic-regression GLM** baseline.

89.76%

ROC AUC — BEST OF ALL MODELS

81.83%

CLASSIFICATION ACCURACY

The simplest model won.

Why does a classical GLM beat sophisticated AI models?

Does this result generalize to the US market?

- We apply the same binary logistic-regression GLM framework to a new dataset: Dow Jones Industrial Average, paired with Reddit WorldNews headlines.
- The DJIA closely tracks the S&P 500 and serves as a valid proxy for the US market.

ORIGINAL PAPER: NIGERIA

NGX All-Share Index

Domain-specific **financial** news

TF-IDF + logistic regression

81.83% accuracy · **89.76%** ROC AUC



OUR STUDY: UNITED STATES

DJIA + Reddit WorldNews (2008 - 2016)

General **world** news headlines

Same TF-IDF + logistic regression

GLM with diagnostics

1,984 trading days of DJIA prices paired with daily Reddit news headlines

1,984

TRADING DAYS (2008 - 2016)

53.6%

GAIN DAYS (Y = 1)

46.4%

LOSS / UNCHANGED (Y = 0)

OUTCOME

- **Gain (y = 1)** if the DJIA closed higher
- **Loss / Unchanged (y = 0)** otherwise

SENTIMENT

- Daily score from the General Inquirer dictionary
- Net positive vs. negative words, bounded -1 to +1

DATA SPLIT

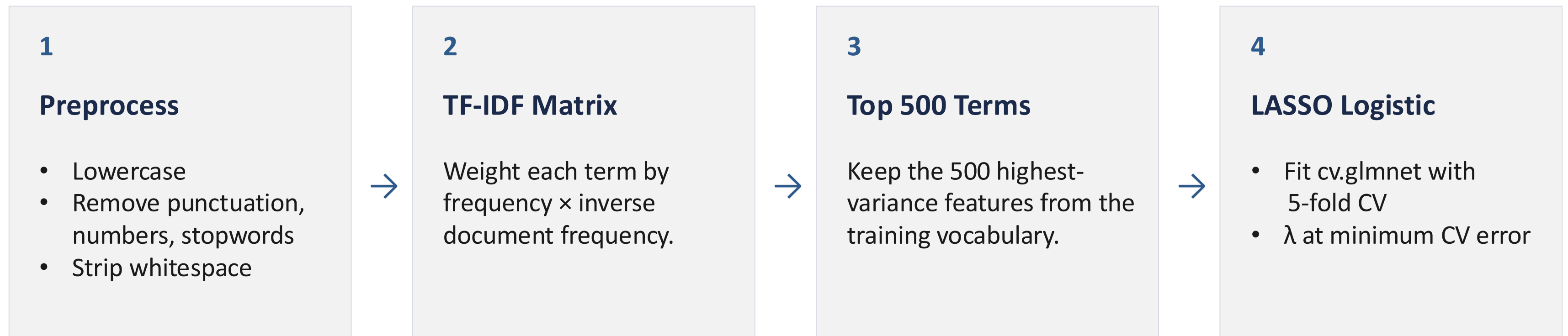
- **70 / 15 / 15** train / validation / test split
- Prevent data leakage

PART I

Replicating the Paper's Method

Run the same TF-IDF + logistic-regression pipeline on US data and test whether the result transfers.

Replicate the paper's TF-IDF pipeline



TF-IDF

- Turns each day's headlines into a numerical vector
- Down-weight common words so the informative ones count more

The method does not transfer: US accuracy collapses to random chance

PAPER: NIGERIA (NGX)

81.83%

89.76% ROC AUC

OURS: U.S. (DJIA)

50.84%

50.00% ROC AUC

- The replicated pipeline **predicted Gain every single day**: recall 100%, ROC AUC 50%
- It found no usable signal and simply defaulted to the majority class

PART II

GLM Analysis

Stepwise AIC keeps only yesterday's information while today's sentiment is dropped

SIX CANDIDATES → TWO RETAINED

×	sentiment_score
×	sentiment_ma5
×	win_rate5
×	sentiment_vol
✓	lagged_sentiment
✓	lagged_label

FINAL MODEL

$$\log \frac{P(Gain_t = 1)}{P(Gain_t = 0)} = 0.349 + 3.068 \cdot Sent_{\{t-1\}} - 0.206 \cdot Label_{\{t-1\}}$$

- We model the log-odds of a Gain as a linear function of predictors.
- Stepwise AIC (both directions) then eliminates four of them
- Lower **AIC = 2735.02** confirms a better fit compared to the full model's **2740.46**

Lagged sentiment is significant but practically negligible

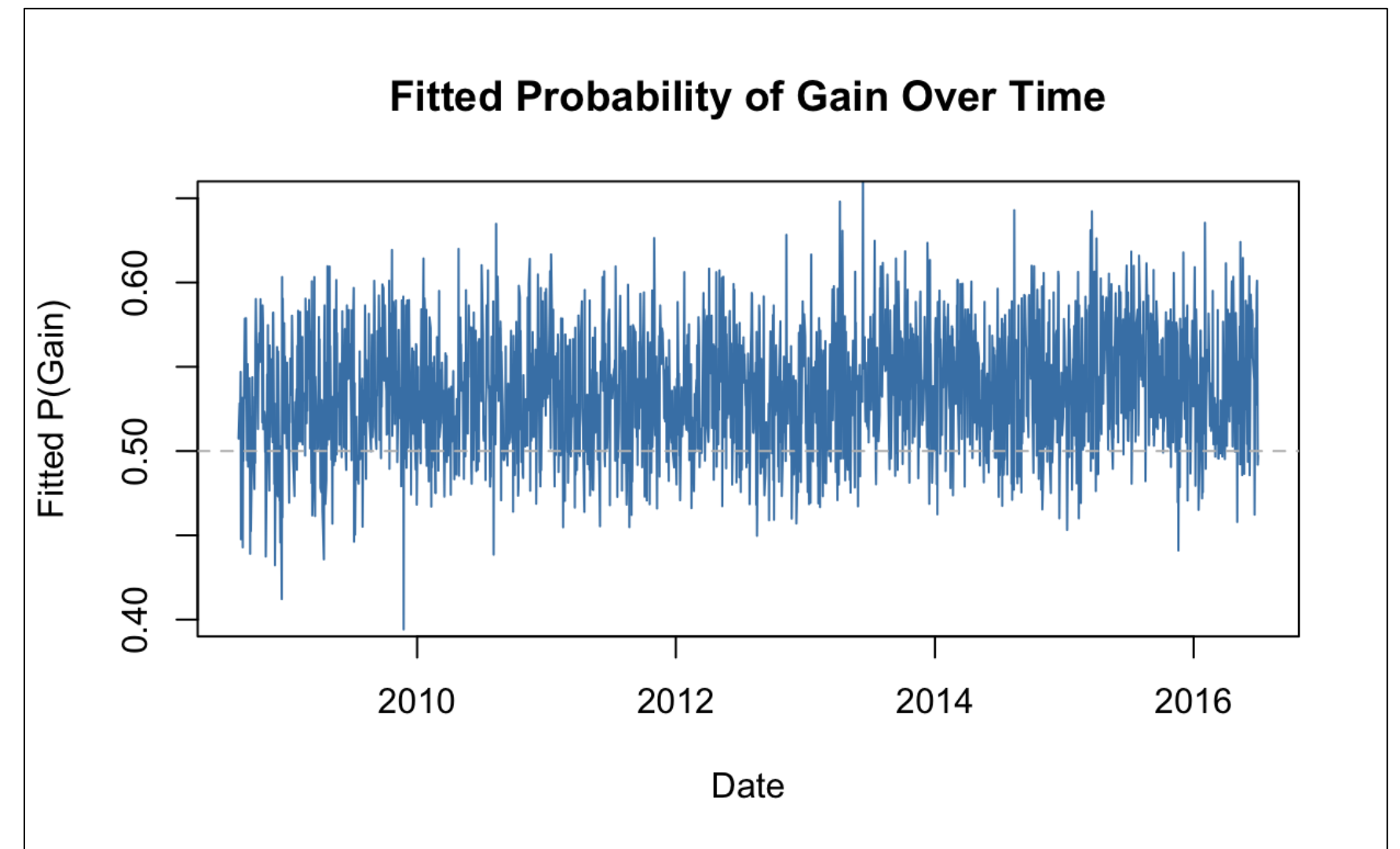
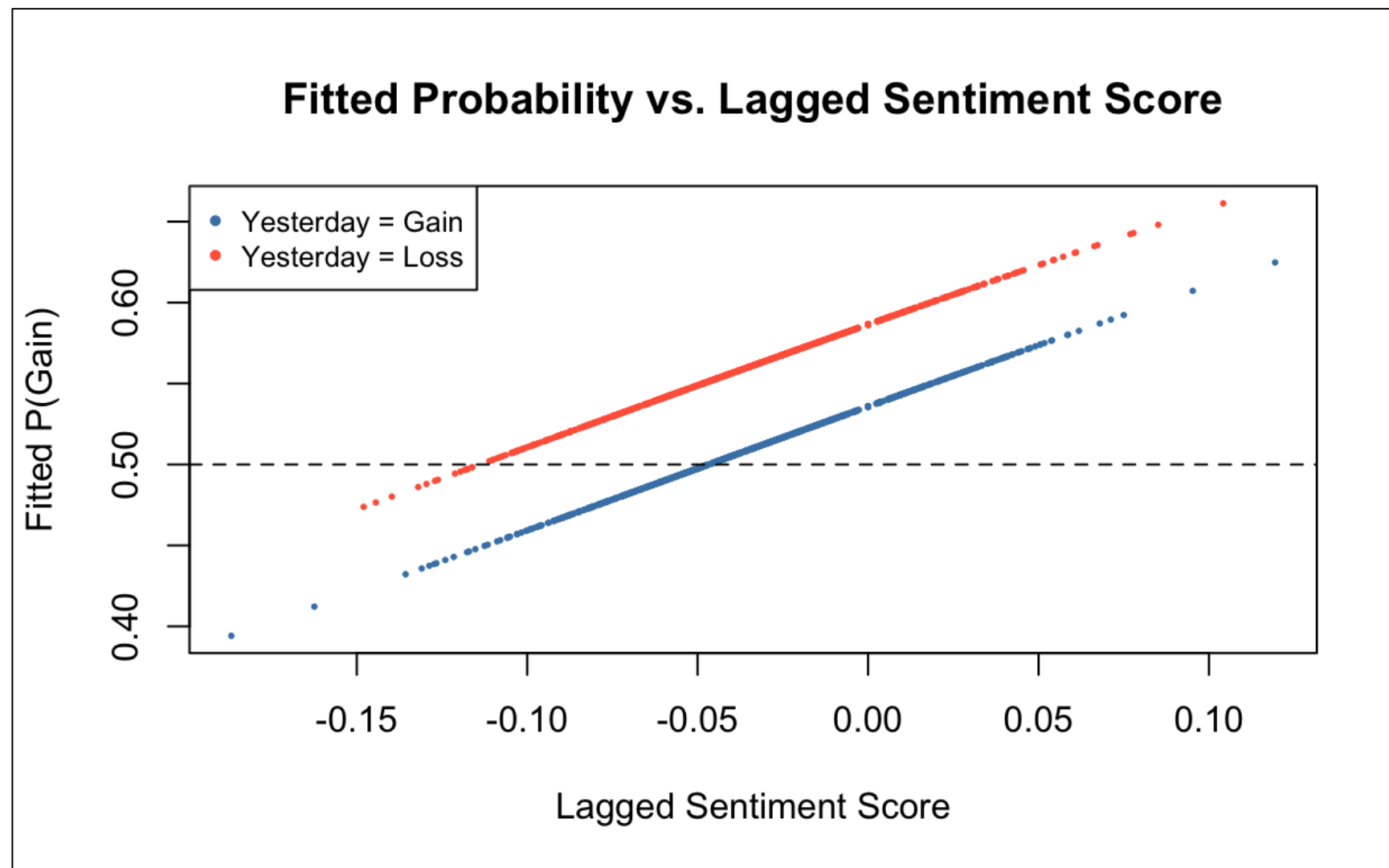
Term	Estimate	Odds Ratio	95% CI	p-value
Intercept	0.349	1.418	—	< 0.001
Lagged Sentiment	3.068	21.50	[1.96, 239.1]	0.012
Lagged Label	−0.206	0.814	[0.681, 0.972]	0.023

An odds ratio of 21.50 sounds large, but sentiment scores span only −0.19 to +0.12, so a one-unit change never occurs in practice.

Likelihood-ratio test: today's sentiment vs. the null model

- $G^2 = 0.925$
- $p = 0.336$
- Today’s sentiment adds no statistically significant predictive power for today’s market direction

Fitted probabilities barely move and stay pinned near 50%

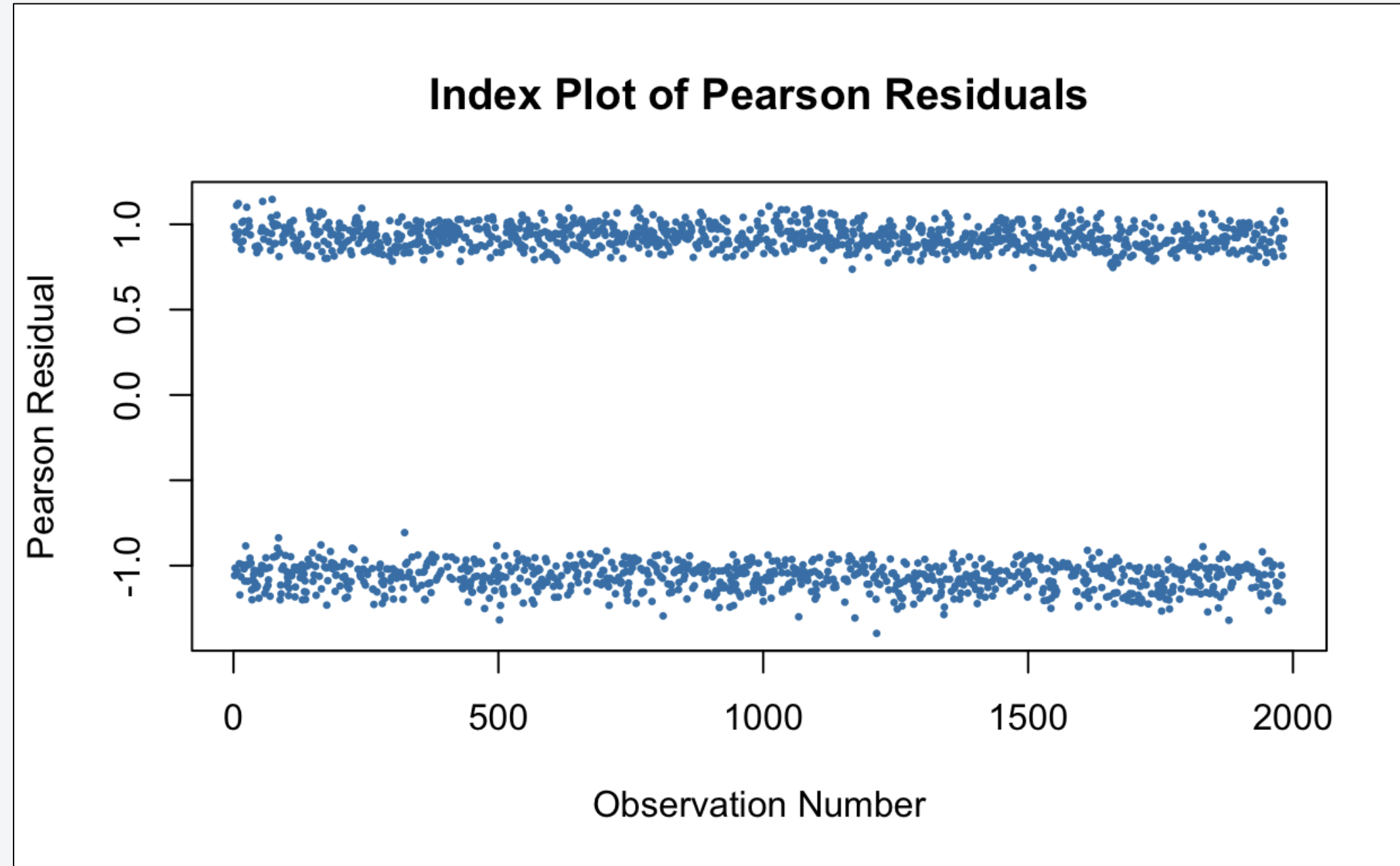


Sentiment shifts $P(\text{Gain})$ by only ~10 points across its whole range

Across 2008 - 2016 the model stays pinned to 0.50–0.55

- *Even through the 2008 financial crisis, the fitted probability of a Gain never departs meaningfully*
- *The GLM cannot tell up days from down days*

The model has good fit, but no predictive power



- *Two clean bands at ± 1 with zero outliers*
- *Goodness-of-fit test $p = 0.477$ (we fail to reject adequate fit)*
- *Good fit is not the same as good prediction: the model passes every check yet barely beats a 50% chance*

54.21% ROC AUC on the DJIA: far below the paper's 89.76%

Model	Accuracy	Precision	Recall	F1	ROC AUC
Shobayo et al. (NGX Nigeria)	81.83%	82.57%	81.15%	81.85%	89.76%
TF-IDF + LASSO (DJIA Reddit)	50.84%	50.84%	100%	67.41%	50.00%
GLM stepwise selected (DJIA Reddit)	53.88%	54.56%	83.35%	65.95%	54.21%

- The DJIA dataset has 53.6% Gain days (“always predict Gain” baseline)
- Both US models sit near the baseline: TF-IDF at 50.84% & GLM at 53.88%
- Neither model demonstrates meaningful lift over random guessing
- The original paper’s 81.83% does not generalize to the US market

CONCLUSION

The Efficient Market Hypothesis explains the null and the GLM confirmed it

The Efficient Market Hypothesis (Fama, 1970) states that in an efficient market, all publicly available information is already reflected in asset prices.
In highly liquid DJIA, algorithms price in headlines within a short amount of time, leaving no exploitable edge for any model.

01

- The paper's TF-IDF method **does not generalize** to the US
- 50.84% vs 81.83% accuracy

02

- Today's sentiment is **not a significant predictor** of DJIA direction (LRT $p = 0.336$)
- An efficient market prices news immediately.

03

- A null result backed by likelihood-ratio tests and clean diagnostics is a **rigorous, meaningful finding.**

Questions?

REFERENCE

Shobayo, O., Adeyemi-Longe, S., Popoola, O., & Ogunleye, B. (2024). Innovative sentiment analysis and prediction of stock price using FinBERT, GPT-4 and logistic regression: A data-driven approach. *Big Data and Cognitive Computing*, 8(11), 143. <https://doi.org/10.3390/bdcc8110143>

Sun, A. (2016). *Daily news for stock market prediction*. Kaggle. <https://www.kaggle.com/datasets/aaron7sun/stocknews>

Dobra, A. (2025). Lecture notes. STAT 424, University of Washington.

Fama, E. F. (1970). Efficient capital markets: A review of empirical work. *Journal of Finance*, 25(2), 383–417.

Friedman, J., Hastie, T., & Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1), 1–22.

Robin, X., et al. (2011). pROC: An open-source package for R and S+ to analyze and compare ROC curves. *BMC Bioinformatics*, 12, 77.

Prollochs, N., Feuerriegel, S., & Neumann, D. (2018). Statistical inferences for polarity identification in natural language. *PLOS ONE*, 13(12), e0209323. <https://doi.org/10.1371/journal.pone.0209323>

Feinerer, I., Hornik, K., & Meyer, D. (2008). Text mining infrastructure in R. *Journal of Statistical Software*, 25(5), 1–54. <https://doi.org/10.18637/jss.v025.i05>

Zeileis, A., & Grothendieck, G. (2005). zoo: S3 infrastructure for regular and irregular time series. *Journal of Statistical Software*, 14(6), 1–27.

Thank You